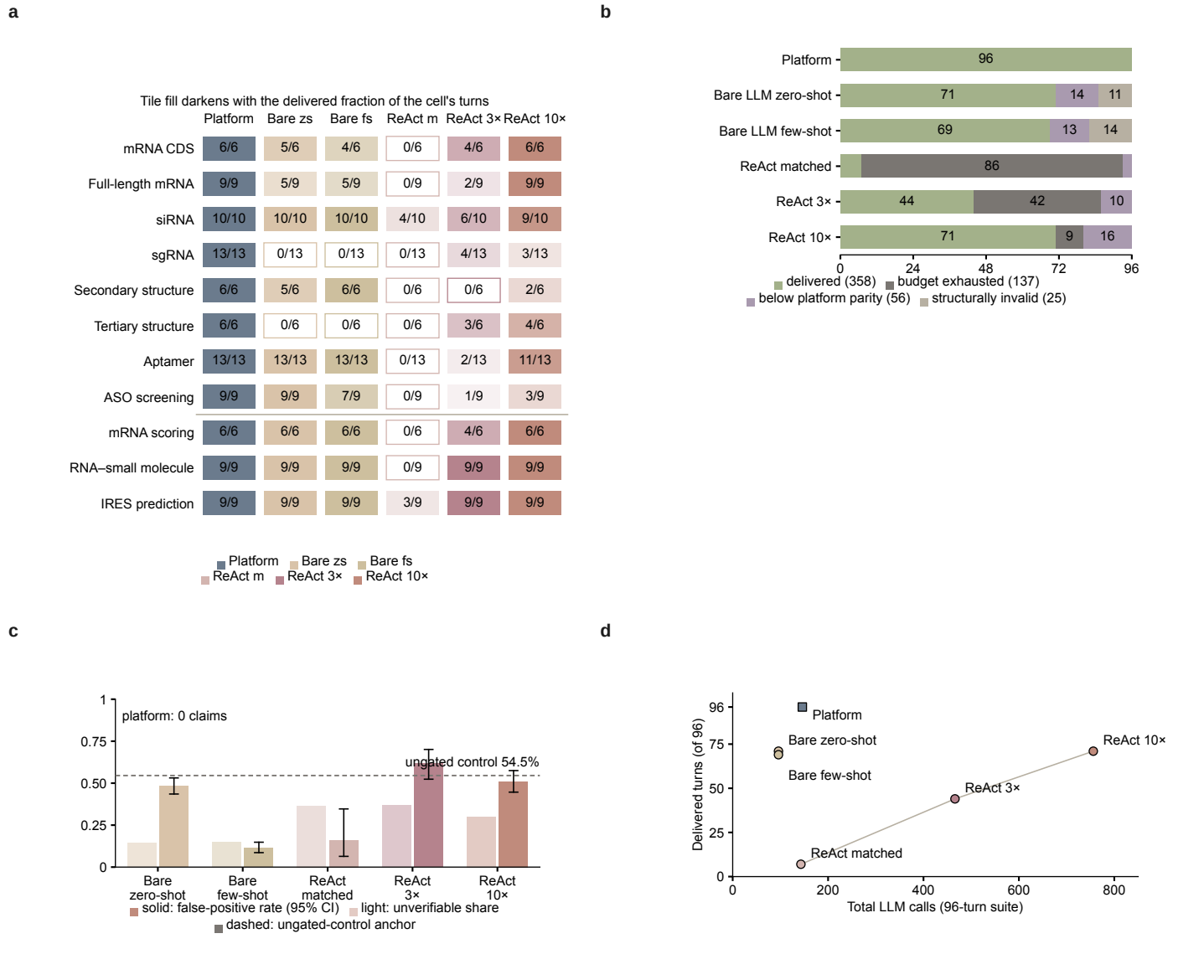

a
Tile fill darkens with the delivered fraction of the cell's turns
Platform
Bare zs
Bare fs
ReAct m
ReAct 3×
ReAct 10×
mRNA CDS 6/6 5/6 4/6 0/6 4/6 6/6
Full-length mRNA 9/9 5/9 5/9 0/9 2/9 9/9
siRNA 10/10 10/10 10/10 4/10 6/10 9/10
sgRNA 13/13 0/13 0/13 0/13 4/13 3/13
Secondary structure 6/6 5/6 6/6 0/6 0/6 2/6
Tertiary structure 6/6 0/6 0/6 0/6 3/6 4/6
Aptamer 13/13 13/13 13/13 0/13 2/13 11/13
ASO screening 9/9 9/9 7/9 0/9 1/9 3/9
mRNA scoring 6/6 6/6 6/6 0/6 4/6 6/6
RNA–small molecule 9/9 9/9 9/9 0/9 9/9 9/9
IRES prediction 9/9 9/9 9/9 3/9 9/9 9/9
b
Platform 96
Bare LLM zero-shot 71 14 11
Bare LLM few-shot 69 13 14
ReAct matched 86
ReAct 3× 44 42 10
ReAct 10× 71 9 16
0 24 48 72 96
delivered (358)
budget exhausted (137)
below platform parity (56)
structurally invalid (25)
c
platform: 0 claims
ungated control 54.5%
1
0.75
0.50
0.25
0
Bare zero-shot
Bare few-shot
ReAct matched
ReAct 3×
ReAct 10×
solid: false-positive rate (95% CI)
light: unverifiable share
dashed: ungated-control anchor
d
Delivered turns (of 96)
96
75
50
25
0
Platform
Bare zero-shot
Bare few-shot
ReAct 10×
ReAct 3×
ReAct matched
0
200
400
600
800
Total LLM calls (96-turn suite)